

Geographic Information Systems · Term Project · Mid-term Review

# Satellites, Night Lights and Words:

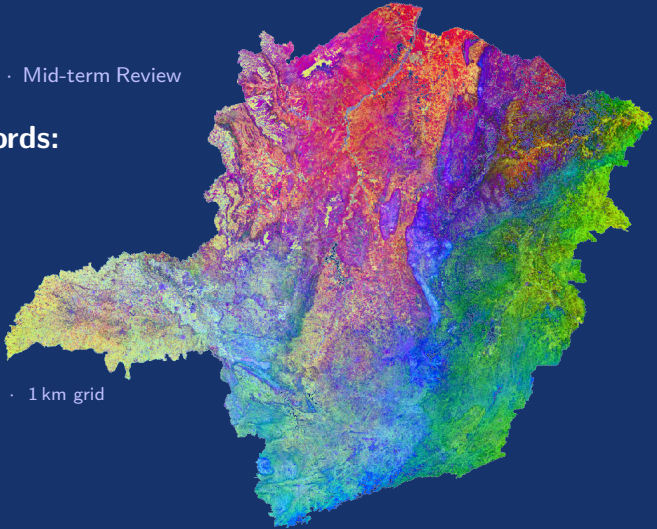
## Estimating Municipal GDP from Satellite Embeddings

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**Hemant** (22CS30029)

Study area: Minas Gerais, Brazil · 853 municipalities · 1 km grid

[github.com/chadsaras/GIS\\_Project](https://github.com/chadsaras/GIS_Project)



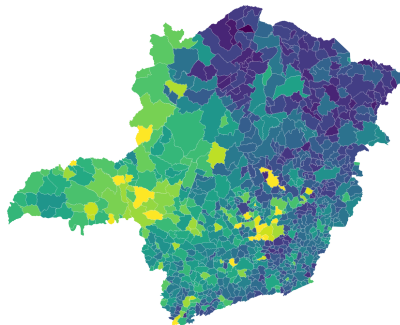
# Problem Statement

## Why Estimate GDP from Space?

- Sub-national GDP is published **late**, often **estimated** from partial data, or **missing** in countries with weak statistical systems.
- Satellites observe every place, every year, with the same sensor — an **independent, consistent** signal of economic activity.
- Inside one Brazilian state, GDP per capita already spans a **50×** range (map): a hard, realistic test.

## Our Goal

Estimate 2022 GDP per capita for each of the **853 municipalities** of Minas Gerais from **free satellite data** — and measure **honestly** how well it works.



Two complementary ideas — never combined, never tested at municipal level

## UrbanCLIP (Yan et al., WWW 2024)

A language model writes a **caption for each satellite tile**; images and captions are aligned contrastively.

GDP gain only **+0.5%  $R^2$** ; random split leaks neighbours; captions checked by hand.

## Satellite Embeddings (Yue, Zhao & Hu, 2026)

Google's **64-number embeddings**, steered by night lights towards economic meaning.

Beats night lights for GDP levels — but only across **countries**; no text, no POI/roads.

## Gaps We Address

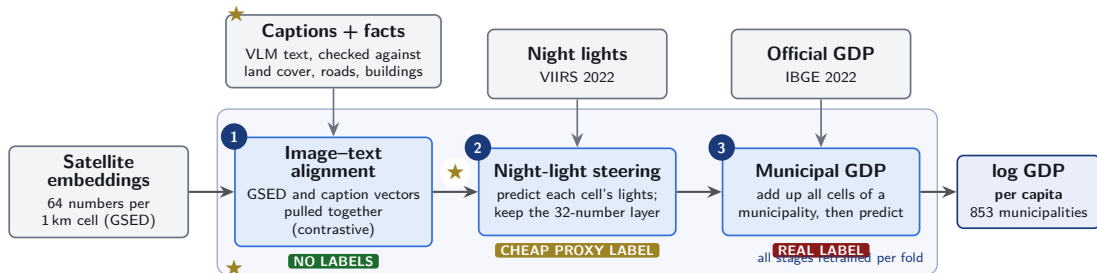
1. The two ideas were **never combined**: do text and steering help each other?
2. Random splits **overstate** accuracy (near-identical neighbours).
3. Hand-checked captions **do not scale**.
4. Country level gives **few training examples**.

## ★ This Project

Both ideas in **one pipeline** · **853 municipalities** · **spatial cross-validation** · **ablations of every ingredient** · **automatic** caption checks

# Our Approach: Three Stages, From No Labels to Official GDP

Supervision gets scarcer and more expensive at each stage



□ input / output

□ learning stage (small networks)

★ our contribution: automatic caption checks, combining A with B, spatial testing

Captions are needed only for training: at prediction time each cell uses its embedding alone. Switching stages off gives the ablations (variants V3–V9).

# Data: Six Free Layers on One 1 km Grid

Every layer averaged onto the same 1 km cells; the GDP target is official statistics, not night lights



**Satellite embeddings**

Google GSED, 10 m → 1 km



**Night lights**

VIIRS 2022



**Built-up land**

ESA WorldCover



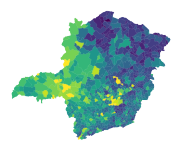
**Roads and POIs**

OpenStreetMap



**Building footprints**

Microsoft + OSM



**GDP and population**

IBGE 2022 (target)

**853**

municipalities

**575k**

1 km cells

**10k**

captioned tiles

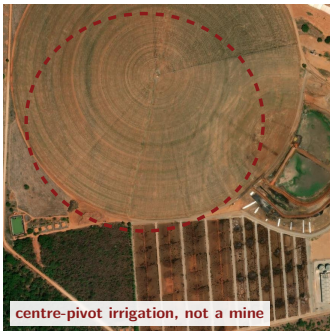
**10M**

building footprints

# ★ Innovation 1: Captions Checked Against the Data

A vision–language model describes each tile; claims the map data contradicts are removed automatically

A real tile from our sample



Esri World Imagery, ~1 m per pixel

Checks use ESA land cover, OSM roads and Microsoft + OSM building footprints. A blind hand audit of 300 captions is in progress.

## Qwen2.5-VL-7B caption, after the checks

- ✗ The satellite image shows a large circular area with concentric rings, likely a mine pit, surrounded by a grid of smaller rectangular plots of farmland.  
**unsupported industry claim**
- ✓ Roads crisscross the area, connecting the plots and the perimeter.
- ✓ Vegetation is sparse, mostly along the edges and in some patches within the farmland.
- ✓ A small body of water, possibly a pond, is visible near the farmland.
- ✓ No buildings or industrial structures are evident, suggesting minimal built-up area.
- ✗ The overall scene appears to be an agricultural landscape with a focus on farming and possibly mining operations.  
**filler**

## What the checks removed

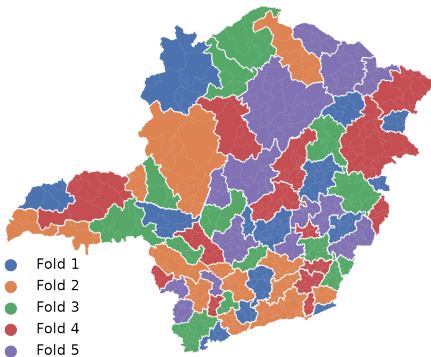
| Reason         | Sentences |
|----------------|-----------|
| Filler         | 4,183     |
| Industry claim | 1,130     |
| Building claim | 1,108     |
| Water claim    | 1,046     |
| Road claim     | 1,039     |
| Farmland claim | 732       |

**Kept 84%**

of 60,023 sentences in 10,000 captions;  
image–text matching removes 246 more  
captions

## ★ Innovation 2: An Honest Test

Whole regions are held out, so no test municipality has a look-alike neighbour in training



### How We Split

1. **70 official regions** of neighbouring municipalities → 5 balanced folds.
2. Each round: 3 folds train, 1 validates, **1 is tested**; **all stages are retrained**.
3. Every municipality is tested **exactly once**, by a model that never saw its region.

### Why It Matters

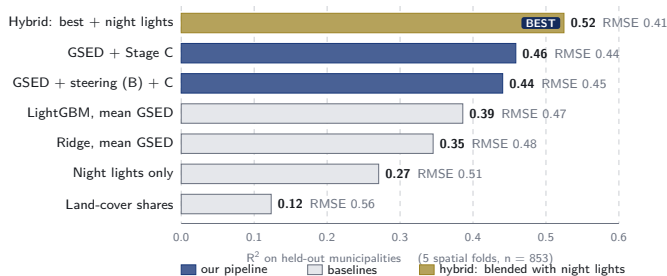
UrbanCLIP split tiles **at random** (60/20/20): near-identical neighbours land in both training and test, inflating the scores.

### Fair Comparisons

Same folds for every model; differences tested with a **block bootstrap** over regions (95% intervals).

# Headline Results: The Hybrid Explains Half of GDP Variation

853 municipalities · 5 spatial folds · every municipality tested once, in a region never seen



# 0.52

R<sup>2</sup> of the hybrid model

RMSE 0.41 · night lights alone: R<sup>2</sup> 0.27

## Research Questions So Far

**RQ1 STEERING** no clear effect ( $\Delta$ RMSE +0.007, 95% CI [-0.028, +0.041])

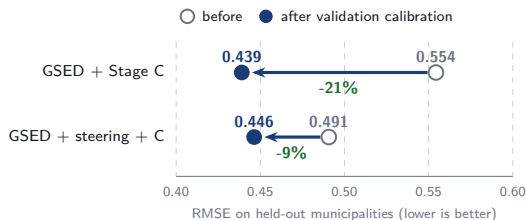
**RQ4 HYBRID** V10 better ( $\Delta$ RMSE -0.028, 95% CI [-0.050, -0.008])

**RQ2-3 TEXT, POI** text variants V5-V9 finishing now

Source: reports/tables/main\_results.csv, rq\_tests.csv. Hybrid = best model blended with the night-light benchmark (weight chosen on validation).

# What We Learned: Calibration Mattered More Than Steering

## Fixing over-confident predictions improved error more than any new ingredient



*Same models and folds before and after; the correction uses validation municipalities only.*

1. **Predictions were over-spread.** Rich places came out too rich, poor too poor (slope 0.6). A straight-line correction fitted on **validation** cut error by ~20%.
2. **Honest re-testing changes conclusions.** Night-light steering looked helpful before the fix; afterwards the gain **vanished**.
3. **Captions need checking.** On featureless tiles the model invents mines, roads and farmland; the data checks remove most of it.

# Progress and Next Steps

About 80% done — the text experiments finish this week

Data, grid and folds



Done

Captions and cleaning



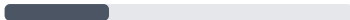
Audit left

Models and experiments



V5–V9 running

Error analysis and maps



Code ready

Report and slides



Next

| Research question            | Status   |
|------------------------------|----------|
| RQ1 Night-light steering     | ANSWERED |
| RQ2 Text (captions)          | RUNNING  |
| RQ3 POIs and roads           | RUNNING  |
| RQ4 Hybrid with night lights | ANSWERED |
| RQ5 Where it fails           | NEXT     |

## Next

text results → caption audit → error analysis and 1 km maps  
→ report

Free satellite embeddings, checked captions and night lights,  
tested honestly on 853 Brazilian municipalities.

**0.52**

$R^2$  of the hybrid  
on unseen regions

**0.27**

$R^2$  of night  
lights alone

**84%**

caption sentences  
kept after checks

**—21%**

error after  
calibration

**Still open:** whether text and POIs add information (RQ2–3), and where the model fails (RQ5).

## Questions?

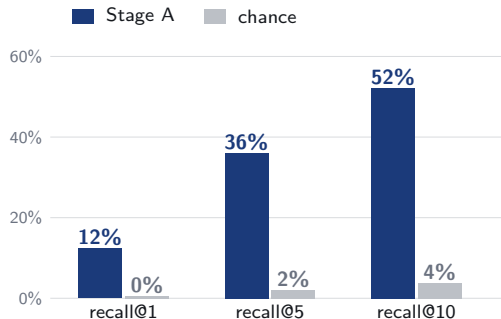
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# Backup: Data Sources

| Layer                | Source                                           | Use                                        |
|----------------------|--------------------------------------------------|--------------------------------------------|
| Satellite embeddings | Google Satellite Embedding V1 (AlphaEarth), 2022 | image input, averaged 10 m → 1 km          |
| Night lights         | VIIRS DNB annual V2.2, 2022                      | Stage B proxy label; night-light benchmark |
| Land cover           | ESA WorldCover v200, 2021                        | masking, fact sentences, caption checks    |
| Roads, POIs          | OpenStreetMap (Geofabrik, 2023-01-01)            | features, fact sentences, caption checks   |
| Buildings            | Microsoft ML footprints + OSM                    | caption checks                             |
| Caption imagery      | Esri World Imagery (ArcGIS Location Platform)    | input to the vision–language model         |
| GDP, population      | IBGE municipal GDP 2022, Census 2022             | target: log GDP per capita                 |

## Backup: Stage A Aligns Embeddings and Captions

For a validation tile, how often is its own caption among the top  $k$  of 256 candidates?



About  $13\times$  **chance** at recall@10. Part of the alignment comes from the fact sentences; variants **V8** (captions only) and **V9** (facts only) separate the two.

## Backup: What the Captions Look Like



**Forest**

A predominantly forested area with patches of cleared land.



**Farmland**

A large, circular area of brown farmland, likely recently plowed, dominating the center.



**Village edge**

A sparsely built area with scattered houses and small buildings, likely residential.



**Town centre**

A densely built-up area with numerous houses featuring red-tiled roofs, indicating a residential neighborhood.

*First sentence of each caption that passed the checks.*